KTH Royal Institute of Technology

School of Electrical Engineering and Computer Science

Opposition Review[1]

Master's Programme in Machine Learning

Author of the thesis	Robert Kihlborg
Title of the thesis	Latent Grid Distributions for Better Robustness in Beat Tracking *Encoding Ambiguity with Metric Distributions over a Learned Tatum Grid*
Name of the opponent	Markus Johnson Swegmark
Date of the opposition	_____

The questions below are a guide. Answers should be justified, and potential issues should be remarked on in enough detail for the author to act on them, either here or as comments in an annotated copy of the thesis.

1 Summary

Briefly summarise the work in one or two paragraphs.

2 Title, Introduction, Background

Do the report title and abstract justly reflect the contents of the report?

Are the project background and underlying interest adequately described?

Is the introduction with general survey of the area clear and complete?

Is the research question, aim, or purpose clear?[2]

3 Theory and Previous Work

Is the theory description sufficient to understand the project, and is it explicitly connected to the project?[3]

Is the description of related work complete enough, and synthesised so as to describe how the project relates to previous research?

[1] Adapted from Kerstin Frenckner's original template.

[2] Since Master's thesis projects differ, it is common, but not compulsory, to include an explicit research question. Regardless, the author should clearly state what the focus of the project is, in the form of a question, aim, or purpose.

[3] The description should be connected to the project, rather than merely being a general description of the area of research.

4 Method

Is the choice of method for tackling the problem justified, including a discussion of alternative methods?

Are the prerequisites for applying the method considered?

Is the method adequately explained?

5 Results, Discussion, and Conclusions

Are the results clearly described?

Is the evaluation performed to assess the results adequate to answer the research question or aim?

Does the author discuss the limitations and possible future work, and compare the achieved results to those of related work?

Are the conclusions credible and supported by the results?

6 Figures, Tables, and Bibliography

Are the figures and tables clear? Consider graphical quality, units, axis labels, and whether each caption is informative enough to make the figure or table self-contained.

Are the figures and tables a valuable support to the main text? Consider whether they are referenced in the text, whether they provide new or more accessible information, and whether any are superfluous.

Is any figure or table necessary for understanding the text missing?

Is the source of information and the origin of graphics clear, for instance when a figure is based on an external reference, and is copyrighted or Creative Commons material handled adequately?

Rate the bibliography in terms of coverage and quality of the references included.

7 Strengths, Weaknesses, and Structure

What are the strong aspects of the work and of the report?

What are the weak aspects of the work and of the report?

Which sections of the report were difficult to understand? Why, and what could be done to improve this?

Other general comments on the report and its structure, including whether there is a logical line of argument from the introduction to the conclusions.

8 Outlook and Concluding Assessment

What is your view of the news value of the work?

Does the report put the project into a larger perspective, considering for example societal impact, sustainability (economic, ecological, or societal), and ethics?

What is your concluding assessment of this Master's thesis project?

9 Seminar Questions

List about six questions to ask the presenter at the seminar.

The questions below are meant to be asked in this order. Each one names the passage it refers to, so that the discussion can stay on a claim, a table, or a design choice in the report.

1. The introduction argues that a distribution over metric interpretations cannot be recovered from framewise beat probabilities, and that a categorical distribution over subgrid states can represent that ambiguity (Figure 1.3 and Section 1.1). Chapter 5 reports that the trained subgrid usually places its mass on a single metric signature, often with probabilities near zero, and that mean squared error was used after binary cross-entropy produced numerical errors. On GTZAN, the clearest gain against other systems without a dynamic Bayesian network is downbeat CMLt (Table 4.5). How much of that gain do you attribute to an explicit distribution over meters, and how much to the penalty on inconsistent neighboring states? What further measurement would separate those two accounts?

2. The first research question asks whether a network can learn a tatum grid when no tatum annotation exists. The grid is trained with soft recall at the annotated beats, which requires those beats to lie on the grid and only mildly prefers a coarser grid. Aggregate recall on the cross-validation set is about 91% (Table 4.1). On JAAH and HJDB, AMLt is higher than recall, which the report reads as a grid that is sometimes slower than the beat, and Figure 4.1c shows a jazz excerpt at 272 bpm on which the grid is too slow. What has been learned in the successful cases: a tatum in the sense of Section 2.1, or a beat grid that is sometimes subdivided? How should the fast-tempo failures enter the answer to the research question?

3. Section 3.2.3 presents Viterbi decoding of the subgrid state sequence as the inference procedure the architecture is built for. On the cross-validation set, thresholding the marginal probabilities gives higher CMLt than Viterbi (Table 4.3). On GTZAN the two decoders are close, with a small advantage for Viterbi (Table 4.5). Recordings longer than 30 seconds are decoded window by window, so Viterbi is never run on one state sequence for the whole piece. How much of the difference between the two decoders do you attribute to that choice? What result would show that the decoder ought to use the full state distribution, beyond the effect the subgrid loss already has on the marginal activations?

4. The learned grid is compared with one predominant-local-pulse configuration, with the window, hop, and tempo range chosen on the combined training and validation data. That baseline is omitted from the GTZAN grid table, because a weak baseline would exaggerate the improvement (Section 4.1.1). Section 1.5 also sets aside an experiment that would pass a non-learned grid into the same subgrid. On GTZAN beat CMLt, the system of Gagneré et al. without a dynamic Bayesian network remains slightly ahead (Table 4.5). Which result do you take the experiments to have established: that the learned grid improves on this pulse baseline, that the full system is competitive with recent beat trackers that do not use a dynamic Bayesian network, or that the split into grid and subgrid is what produces the downbeat continuity gain?

5. A reason given for avoiding dynamic Bayesian network postprocessing is that the allowed tempos, time signatures, and meter changes leave some music out (Section 1.1). In Table 4.4, constant 4/4 accounts for about three quarters of the cross-validation pieces that have downbeat labels, and CMLt falls for every system once the bar length is not constantly four beats. Figure 4.2 shows the subgrid shifting probability from one phase to another across a bar of 2/4, described in the text as a slow change of confidence. On the music that a typical dynamic Bayesian network handles poorly, where does this model improve on that postprocessing, and where do the training distribution and the transition penalty reproduce a similar preference for a stable 4/4?

6. The abstract notes that the aggregate scores can look strong while inspection of individual excerpts still shows failures on comparatively straightforward material. Chapter 6 describes the system both as a proof of concept at the level of beat-activation models and as a design that might remain a dead end. Taking the downbeat CMLt numbers in Table 4.5 as given, which behavior in Figures 4.1b, 4.1c, and 4.2 is, in your view, the clearest sign that the model is not yet doing what the grid and the subgrid were designed to do? What change would you try first?

A Instructions to the opponent

The aim of the opposition review is twofold. As an opponent, one should demonstrate the learning outcome of being able to reflect on, and critically review, others' scientific achievements. For the presenter, the feedback is an opportunity to improve the report further, and to demonstrate an ability to address constructive criticism by considering the remarks when preparing the final version of the thesis.

The review should therefore identify strengths of the work and, in relative detail, suggest improvements that would make the report better in completeness, ease of understanding, and readability. The evaluation questions above are a guide. A different outline is acceptable, provided the review contains enough remarks for the presenter to consider the suggested improvements. Details may instead be given as comments in an annotated PDF of the report. If so, state briefly in the relevant section where those comments are provided.